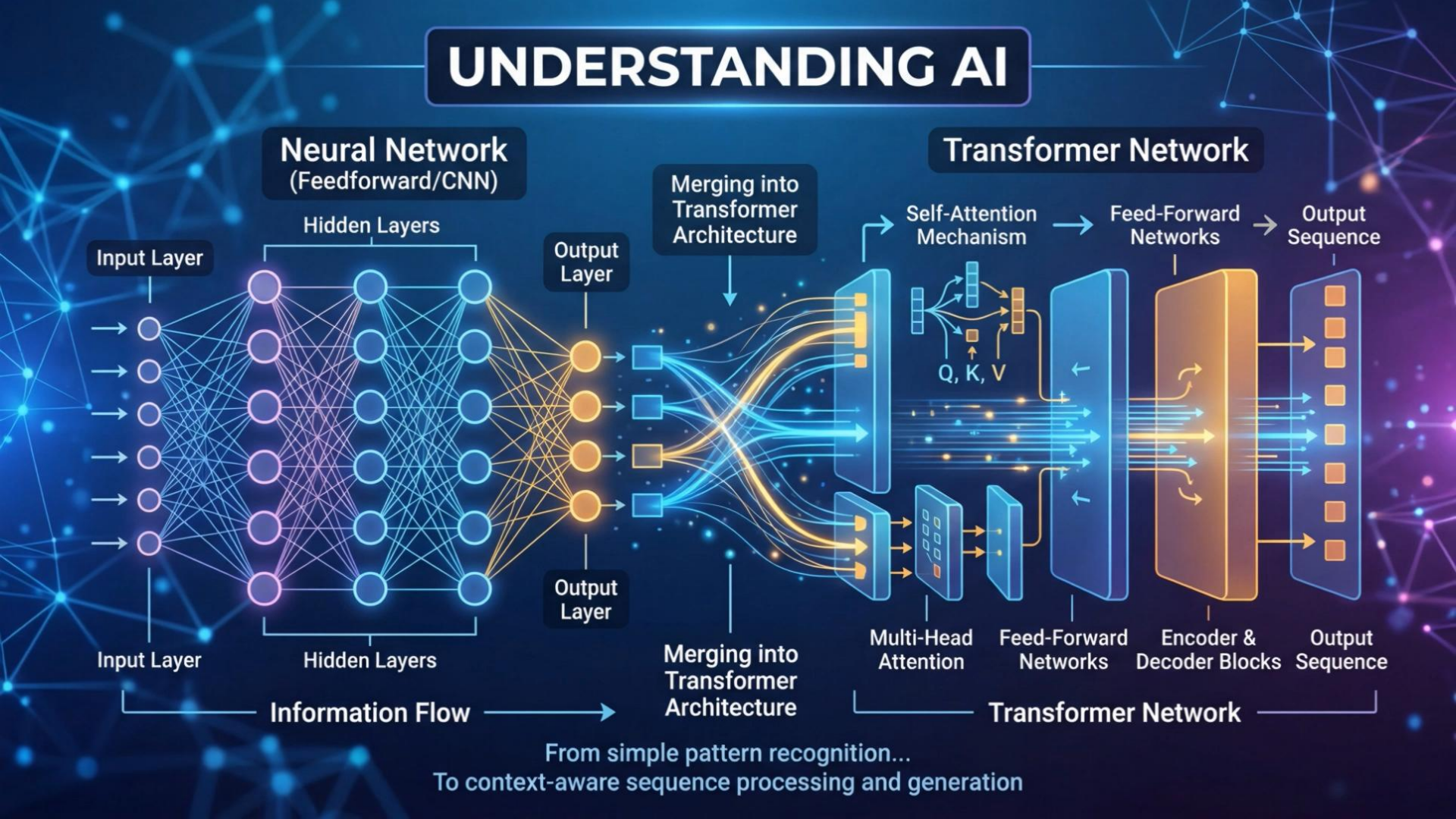
UNDERSTANDING AI



What We'll Cover Today

1

What Is AI?

Neural networks, key terminology, costs, shortcomings & bias

2

The cost of AI

Monetary cost of implementing an AI Data center.

3

Concerns About AI

Learning, cost, youth anxiety, cultural representation

4

Using AI for Research

How to ask good questions and avoid biased answers

5

AI Resources

Where can you find useful ideas.

huggingface.co

6

Fun Stuff

What anyone can do with AI.

PART 1

What Is AI?

Clearing the fog around a big buzzword

AI Is a Broad Category

It is because we have a broad definition of intelligence.

THE FOUNDATION SPECTRUM

Simple Rules	Machine Learning	Neural Networks	Deep Learning
If/then logic, subsumption architecture, basic sorting	Learns patterns from data	Brain-inspired computing	Many layered neural nets

MODERN AI APPLICATIONS (all built on deep learning)

Large Language Models (LLMs)	Generators & Denoisers	Vision & Audio Recognition
ChatGPT, Claude, Gemini, trained on vast text to generate, summarize, and answer	Image generators (DALL·E, Midjourney), music AI, learn to create from noise	Facial recognition, medical imaging, speech-to-text, sound classification

Not everyone agrees on what deserves the label "AI." The Turing Test was proposed in 1950s. Our current LLMs have passed this test. But do they understand?

Neural networks date back to the 1940s, what changed is our computing power and data.

How Neural Networks Actually Learn

Think of it like teaching a child to recognize dogs:

1. **Show many examples**

   Thousands of labeled photos — "this is a dog," "this is not a dog"

2. **Get it wrong — adjust**

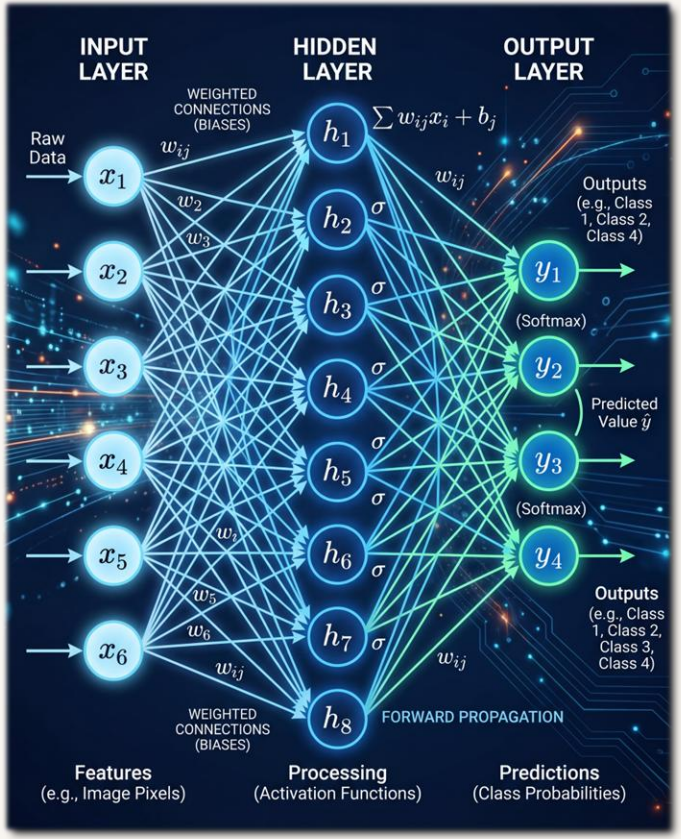
   Each mistake slightly updates how the network processes images

3. **Get it right — reinforce**

   Correct answers strengthen those pathways, wrong ones weaken them

4. **Repeat — millions of times**

   After enough examples, patterns emerge and generalize to new images



Do We Know How These Actually Work?

Yes. But it takes enormous effort to find out the details.

Large Language Models (LLMs)

ChatGPT, Claude, Gemini

Generators & Denoisers

DALL·E, Midjourney, Suno

Vision & Audio Recognition

Facial recognition, speech-to-text

Teaching any of these models require “good” data. Training data requires an input and an expected output.

Example:
Input: The sky is blue because it is a reflecting the oceans.
Output: False, it is blue due to blue wavelengths being scattered in the atmosphere due to dust, water vapor and other small particles….

3Blue1Brown

Sample clip of LLM training data.


```
[
{
"from": "human",
"value": "Augment this coding dilemma to handle not just tuples laden with duos, repetitions and idiosyncratic numbers of elements, but to ..."
},
{
"from": "gpt",
"value": "For implementing a function to calculate the median of a list without using built-in functions or sorting, we would have to implement our approach to locate the middle element. ..."
}
]
```

HOW STABLE DIFFUSION TAKES RANDOM NOISE TO A FULL DEPICTION

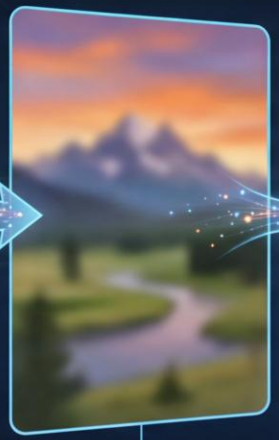
INPUT: RANDOM NOISE

PURE GAUSSIAN NOISE (START)

EARLY NOISE REDUCTION

Step 1: INITIAL STRUCTURE EMERGES

REFINING COARSE DETAILS



Step 2: BUILDING MAIN FEATURES

MID-PROCESS REFINEMENT

Step 3: ADDING TEXTURE & DEFINITION

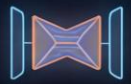
LATE-STAGE SHARPENING

Step 4: FINE-TUNING DETAILS

FINAL DEPICTION

FINAL RESULT: DETAILED DEPICTION (High Quality)

ITERATIVE DENOISING PROCESS



U-NET NETWORK

TEXT CONDITIONING

PART 2

The cost of AI

Monetary cost of implementing an AI Data center.

The Cost of Teaching a Neural Network

Training large AI models is extraordinarily expensive — in dollars, energy, water and time

~$4M

Cost to train GPT-3

(2020)

$100M+

Estimated GPT-4 training cost

(2023)

Months

Time for a single training run

on thousands of GPUs

Once

How often we can afford to retrain from scratch

Existing models get updated, not rebuilt

The Cost of Refining a Model

Models have been developed from core ones with different purposes. In many cases there is no need to "tune" a model.

~$1,000

Small data set of a 2000 inputs and example outputs

Here quality matters and the cost that is not reflected is the man-power to get quality data.

$10,000

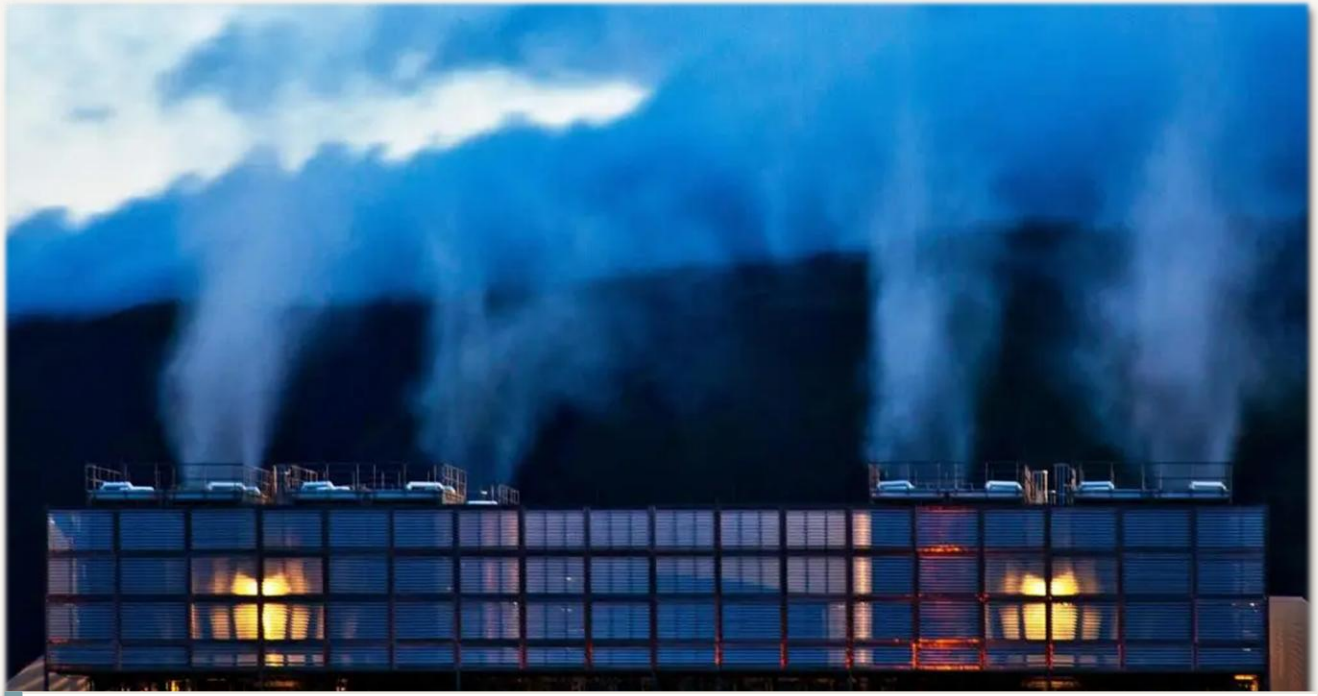
Estimated 5 days to train 100K sets

You need good quality.

Tuning/Refining

You can now stitch several models together for different tasks where some can feature as summery agents, others as coding agents. If there isn't a general model with the capabilities you need, you can now train an existing model by using high quality data and training it further.

Data Center Costs: Water

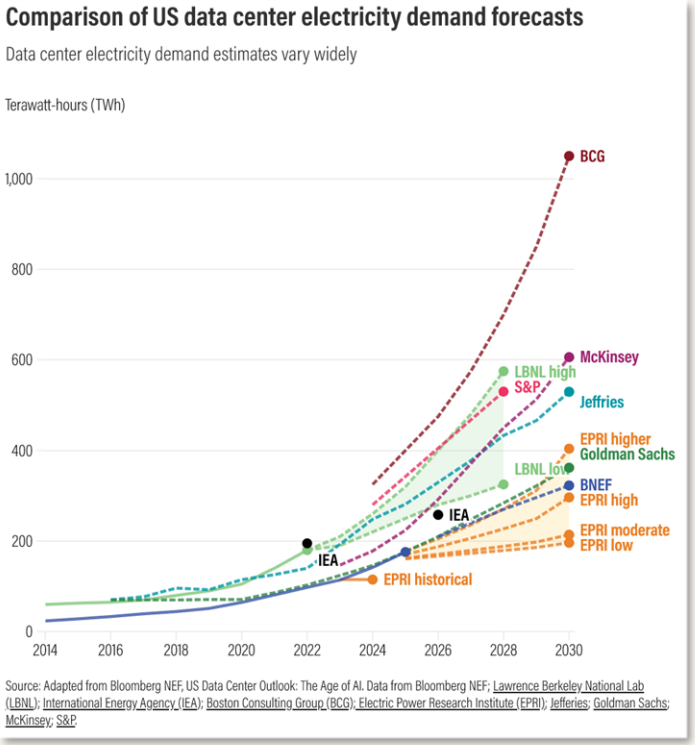


Source: Google. Data center in The Dalles, Oregon releases water vapor and steam from cooling towers

Cooling towers that use water provide a 20% to 50% decrease in energy consumption.

Data center Costs: Power

- AI is part of our future.
- But in what form?
- But the competition is stiff and there can be many losers in this!



Source: World Resource Institute.

- Who pays to build the new power plants?
- If the AI market crashes, who carries the bill for the power plants?

Project Matador – Data center

- Carson County, Texas
- 5,855 acre project
- 11-17 Gigawatt power from: Nuclear, Natural gas, Solar
- 2.5 million gallons of water a day from the aquafer.
- Why Texas? Least regulated.
- Loud!?

Carson County, Texas

- Average Annual Rainfall: ~20.92 inches.
- Rainfall Characteristics: Short, intense showers are common, with rainy periods usually lasting 1–2 days.
- Growing Season: Roughly 191 days.
- Climate Context: The area is located in the Texas Panhandle, which is known for a semi-arid to sub-humid climate, drier than the eastern part of the state.

Data center Costs: Consumer Electronics

AI Data centers need the fastest memory available. The demand is extremely high!
We as consumers would never need these high-speed memory in our electronics.
But the same factories that make the fast memory also make the consumer memory. They make more money by making the high-speed memory chips! This applies to other components as well!

Scalpers are seeing the trend and buying up memory and selling it at premium prices.

Scammers are also helping themselves, by offering too good to be true deals!

The pricing future is as uncertain as the AI market is!

G.SKILL 8GB DDR4 3200 (PC4 25600) Desktop Memory Model F4-3200C16S-8GVKB

- CAS Latency: CL38
- Voltage: 1.35V
- Timing: 16-18-18-38
- Recommend Use: AMD X570 / X470 / B550 / B450 Intel Z790 DDR4 / Z690 DDR4 / Z590 / Z490 / Z390 / Z370 / Z270 / Z170
- Model #: F4-3200C16S-8GVKB

~~$76.99~~
$69.99
Save: $7.00 (9%)
More options from $25.84 - $99.20
FREE SHIPPING from United States
ADD TO CART
Compare

Total Micro 8GB DDR4 SDRAM Memory Module - For Desktop PC, Workstation - 8 GB (1 x 8GB) - DDR4-3200/PC4-25600 DDR4 SDRAM - 3200 MHz Single-rank Memory - CL22 - 1.20 V - Non-ECC - Unbuffered ...

- Brand: Total Micro
- Capacity: 8GB
- Speed: DDR4 3200 (PC4 25600)
- Part Number: AB120718-TM
- Model #: AB120718-TM

$82.99
More options from $80.56 - $83.74
FREE SHIPPING from United States
ADD TO CART
Compare

G.SKILL Flare X5 8GB 288-Pin PC RAM DDR5 6000 (PC5 48000) Desktop Memory Model F5-6000J3038F8GH1-FX5

- CAS Latency: CL30
- Voltage: 1.35V
- Timing: 30-38-38-96
- Brand: G.SKILL
- Model #: F5-6000J3038F8GH1-FX5

~~$142.99~~
$129.99
Save: $13.00 (9%)
FREE SHIPPING from United States
ADD TO CART
Compare

(64)

Patriot Viper Elite 5 16GB 288-Pin PC RAM DDR5 6000 CL30(PC5 48000) Hynix Desktop Memory Model VEB516G6030W Compatible with AMD / Intel Gen14 Only

- CAS Latency: CL30
- Voltage: 1.35V
- Timing: 30-40-40-76
- Brand: Patriot
- Model #: VEB516G6030W

~~$219.99~~
$199.99
Save: $20.00 (9%)
More options from $46.99 - $226.32
FREE SHIPPING from United States
ADD TO CART
Compare

PART 3

Concerns About AI

Five things we should all be thinking carefully about

Why Aren't Neural Networks Doing Everything?

Hallucinations

AI confidently states false information. It doesn't know what it doesn't know, it predicts the most plausible-sounding next concept, not the truest one.

Lack of Heuristics

Heuristics are learned behaviors of creatures that require very little effort. Consider how much effort it might take to do a new task. After a while it becomes easier. This is Heuristics or in some versions common sense.

Context Limitations

AI Company CEOs believe bigger models will solve this. Researchers disagree! Newer models have massive Context Windows but they come at a massive computational cost. This new massive context can have a new issue of “lost in thought” problem.

Fragility

Tiny changes in wording can produce wildly different answers. The same question phrased differently may get opposite responses. Confident Sycophant Responses make them appear reliable, and only later causes serious harm due undeserved trust.

Energy Demands

Running AI inference (asking it questions) at scale consumes enormous power. Practical limits exist on how widely these systems can be deployed.

Backward Propagation

The most common way to have AI learn is to pass data through an AI Neural Network. Take outputs and calculate the error. Pause the network and run in the reverse to update correction. Humans learn on the fly (as we go in real time).

Why Aren't Neural Networks Doing Everything?

The Real-world is complex!

It is easy to take billions of digital books and feed it to a computer AI model to learn. It is harder to get that same quality of data from the real-world movements. The more complex the task, the more data you need.

Concerns!

- Ethical concerns of how the data is collected.
- What is the risks of real-world AI interactions?
- (Plugin your favorite Sci-Fi movie in here)
- Who reaps the rewards!?

Solutions?

We will also need different approaches and models to address these issues.

But we also need more data!
And what will that form be?

Possible Benefits!

- In home care?
- Un-bias Policing?
- Dangerous Tasks?
- Loading the dishwasher?
- The list is unending!

DeepMind robotics lab tour with Hannah Fry

Bias in Neural Networks

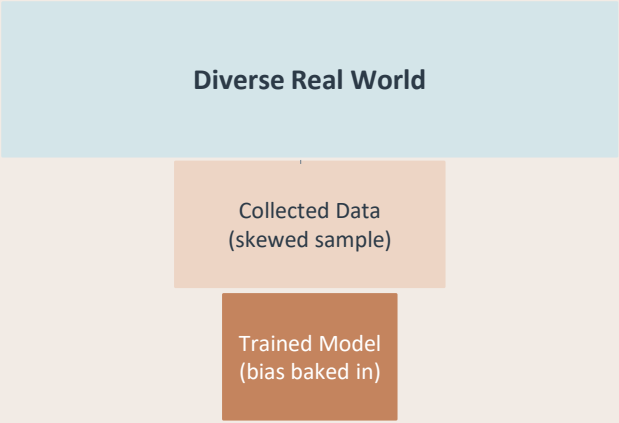
"Garbage in — garbage out"

AI learns from data created by humans, and humans carry biases. If the data used to train a model over-represents certain groups, languages, cultures, or viewpoints, those biases become baked into the model.

This isn't a bug that can be easily patched. It is a **structural feature** of how these systems are built.

Real-world examples:

- Most training data is in English — non-English speakers get worse performance
- Racial and gender biases in facial recognition and hiring tools are well-documented
- Rare conditions, minority cultures, and marginalized voices are underrepresented
- Our current models are generated by using literature online and books. Who is not participating in generating this content and is that causing a bias.



AI output reflects the bias of training data — not the full human experience

Productivity vs. Learning

We can do more — but are we learning more?

AI dramatically increases what we can produce in a day. A report that once took a week can now be drafted in hours. But there's a question worth sitting with:

If AI did the thinking, did you learn anything?

- Expertise develops through struggle, when AI eliminates the struggle, it may eliminate the learning.
- “First comes learning then understanding.”
- Clinical judgment comes from accumulated experience. Outsourcing thinking to AI may erode that accumulation.
- Many practitioners report a sense of panic when they realize the work is done but they can't explain it or defend it.
- This is especially critical when AI handles complex, nuanced work in human services fields.

Young People

A generation seeing their dreams disrupted.

Social Influencers are the new dream job: because that might be all they see!

Learning does not have to be about a future job. It might be worth learning just to know things you didn't before. Maybe it could have some use to you, but without trying you won't know.

Security

If AI is everywhere then it will be a target for any nefarious actors.

Using sensitive data in LLM

If sensitive data is used in an LLM, it is not automatically incorporated. But a later iteration of the LLM might contain it. Companies that have trained LLMs have used data without asking permission in the past.

Computer System Security

With AI coding and doing quite well, could vulnerabilities be introduced that does not become obvious until too late? Social Engineering is now AI Red Teaming.

Surveillance

AI cameras and online presence could extend further than ever for companies or countries having an AI profile on everyone.

AI in warfare and government use

AI is now part of warfare and governments want their use expanded. Are we ready for what that means?

The positive side: Human Brain vs. Neural Networks: The Numbers

	Human Brain	Large AI Model
"Neurons" / Parameters	~86 billion neurons	~1 trillion parameters (GPT-4 est.)
Power consumption	~12 watts	~1,500 watts (inference server)
Efficiency	7 billion neurons per watt	~1 billion parameters per watt
Learning from examples	Learns a concept from 1-5 examples	Needs thousands to millions of examples
Energy cost to "learn"	12W running all day, every day	Millions of dollars, months of compute
Generalizes to new tasks	Readily — without retraining	Poorly — often fails outside training domain

PART 4

Using AI for Research

How to ask the right questions — and spot the wrong answers

Step 1: Start with a Clear, Open Question

The Goal: Genuine Curiosity

A good research question is open-ended.
You want to understand a topic, not validate what you already believe.

✓ GOOD EXAMPLE

"Why is the sky blue?"

This question is neutral. It doesn't assume an answer.

The LLM models make money by being of use to people. It will try to be considerate of your effort and can lead you down to the wrong conclusions if you give it the “bad prompt”.

Key Principle: Your question should invite learning, not confirm what you already think.

Open question = open mind

Step 2: Don't Bake In Your Conclusion

Leading questions lead to biased answers

✗ PROBLEMATIC EXAMPLE

"How does the blue ocean cause the sky to be blue?"

This question contains a false premise!

Open Question

"Why is the sky blue?"

AI explores all causes, may mention Rayleigh scattering, light wavelengths, atmosphere. Complete picture.

Leading Question

"How does the ocean make sky blue?"

AI may construct a supportive narrative for a false premise, confidently providing bad information.

Step 3: If You Must Test a Conclusion: Falsify It First

When your research question involves a claim or theory, deliberately ask AI to argue the opposite

1

Your Hypothesis

"Does the blue ocean make the sky blue?"

You suspect a connection between two phenomena

→

2

Ask for the Counter-Argument

"Why does the blue ocean have NO bearing on why the sky is blue?"

Ask AI to explain why your hypothesis might be wrong

→

3

Now You're Informed

Read both sides — then form your conclusion

You understand the strongest arguments for AND against

This technique is called falsification, the cornerstone of scientific thinking. If an idea can't survive being challenged, it probably isn't true.

AI Research — Final steps

1. Ask for sources.

The latest models can provide you with sources, such as links to scientific journals and websites. As always make sure the websites are legitimate.

2. Ask what the conclusions are based on?

You might not be aware what is in the context window of the AI model, or you missed bad data in a document you provided. The LLM might have been trained to know when data is X then correctly respond with Y. But your data or the context window has something like X and now it confidently responds with Y. Ask it what the key factors are, on which it based its conclusion, and then verify if those factors are true and if the conclusion follows.

3. Ask how can this be tested and verified. Consider the falsified test!

If AI reaches a conclusion, it could be a good idea to ask it for the falsified position on that conclusion. There might be several falsified positions. Next ask it how you could test these falsified positions to know if the conclusion is true.

PART 5

AI Resources

Huggingface.co

AI Resources: huggingface.co

How to run things locally and “free”.

How the website is used!

People use Hugging Face and its platform Hugging Face Hub to discover, download, and share machine learning models, datasets, and AI demos. Developers and researchers commonly use it to quickly integrate pre-trained models (like NLP, vision, or speech models) into their applications and collaborate with the AI community.

LLM on your machine: LM Studio

People use LM Studio to run large language models locally on their own computer without needing cloud APIs. Developers and hobbyists use it to download models, chat with them, and integrate them into local apps while keeping data private.

Diffusion: ComfyUI and DiffusionBee

People use these tools to install, manage, and run different Stable Diffusion tools and models from a single interface on their computer. It helps developers and artists organize AI image-generation packages, update them easily, and generate images locally without complex setup.

Python: Powerful Script Language

Most LLMs can easily code in Python script and even create very capable applications. The reason is that the libraries they use is incredibly powerful and well supported. In a 5 lines of code you can recreate a simplified LLM.

PART 6

Fun Stuff

Create Music, Images, Websites and and and…

Fun Stuff: LLM ~ $20/mo

Large Language Models. ChatGPT, Claude, Copilot and Gemini!

ChatGPT - OpenAI

1) Very good at all general tasks including coding and reports.
2) Has a built-in image generator.
3) Questionable Upper Management: CEO

Claude - Anthropic

1) Very good at all general tasks including coding and reports.
2) No Image generator.
3) Very good at creating code.
4) Run out of tokens fast with top model! ☹
5) Considered to have an ethical approach to AI.

Gemini - Google

1) Very good at all general tasks with a wide range of image, video, text and code capable features.
2) Google has several models that fall under Gemini and they have been in the business of “AI” for quite sometime.
3) Not sure where their ethics are in usage of data collection?

LLama – Meta/Facebook

1) Massive models that are downloadable.
2) Specialized tasks from scout work to generalized models.
3) Trainable but require expensive equipment.
4) CEO is a creep?

Fun Stuff: Generators and Tools

Images, Video, Worlds, Music!

SUNO.com - Music

1) Amazing and creating professional sounding music.
2) Blending themes and styles with vocals.
3) $10/mo.

Openart.ai - Image, Video clips

1) Best viewed as a model hosting service.
2) Adds new features on a regular basis.
3) $14/mo

Ultralytics.com - computer vision

1) Developed all types of models for computer visions.
2) If you know how to use basic python everything is free to use. Can use a graphics card to refine.

Google Colab - colab.research.google.com

1) Free to run small training tasks in a workbook style.
2) Coding language is in Python. LLMs can help code for you!
3) Free or start at $9.99/mo for Powerful system access.
4) Access to start to explore the bigger side of AI.

Examples

Thank You

Questions & Discussion